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ORIGINAL SCIENTIFIC PAPER
Biomarkers

A new 0 and 2 hour risk assessment for acute myocardial infarction in emergency department patients using a high sensitivity point of care troponin assay

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Aims

A 0 and 2 h strategy using novel point-of-care high-sensitivity troponin I (POC hs-cTnI) measurements may provide rapid and accurate risk stratification for acute myocardial infarction (AMI) in the emergency department (ED) by reducing time to troponin results and improving clinical decision-making.

Methods and results

A prospective multicentre diagnostic accuracy study enrolled 1021 patients presenting to EDs with symptoms suggestive of AMI. POC hs-cTnI measurements were obtained from stored plasma samples at presentation (0 h) and 2 h later using the Abbott i-STAT® hs-TnI assay and compared with central laboratory assay results. A risk stratification algorithm was derived. The primary endpoint was AMI (type 1 and 2) or cardiac death during the index presentation. The primary safety endpoint was 30 day major adverse cardiac events (MACE), incorporating AMI and cardiac death. AMI was diagnosed in 80 patients (7.8%). When identifying low-risk patients, the 0/2 h strategy using the POC hs-cTnI assay demonstrated 98.8% sensitivity (95% CI: 93.2–100%) with an NPV of 99.9% (95% CI: 99.2–100%) for the primary endpoint. For high-risk patients, specificity was 97.7% (95% CI: 96.5–98.5) and PPV 74.7% (95% CI: 64.3–83.4%) for the primary endpoint. The protocol safely identified 66.7% of patients as low-risk within 2 h, with 6 low risk patients (0.9%) having a MACE diagnosis at 30 day follow-up.

Conclusion

A 0/2 h protocol using the Abbott i-STAT® hs-TnI assay safely risk stratified emergency patients with suspected ACS and had comparable performance to two central laboratory-based assays. The use of POC testing supports timely results and may improve the time to clinical decision-making.

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Institutions where work was performed: Royal Brisbane and Women's Hospital, Gold Coast University Hospital, Princess Alexandra Hospital, Logan Hospital, and Townsville Hospital.
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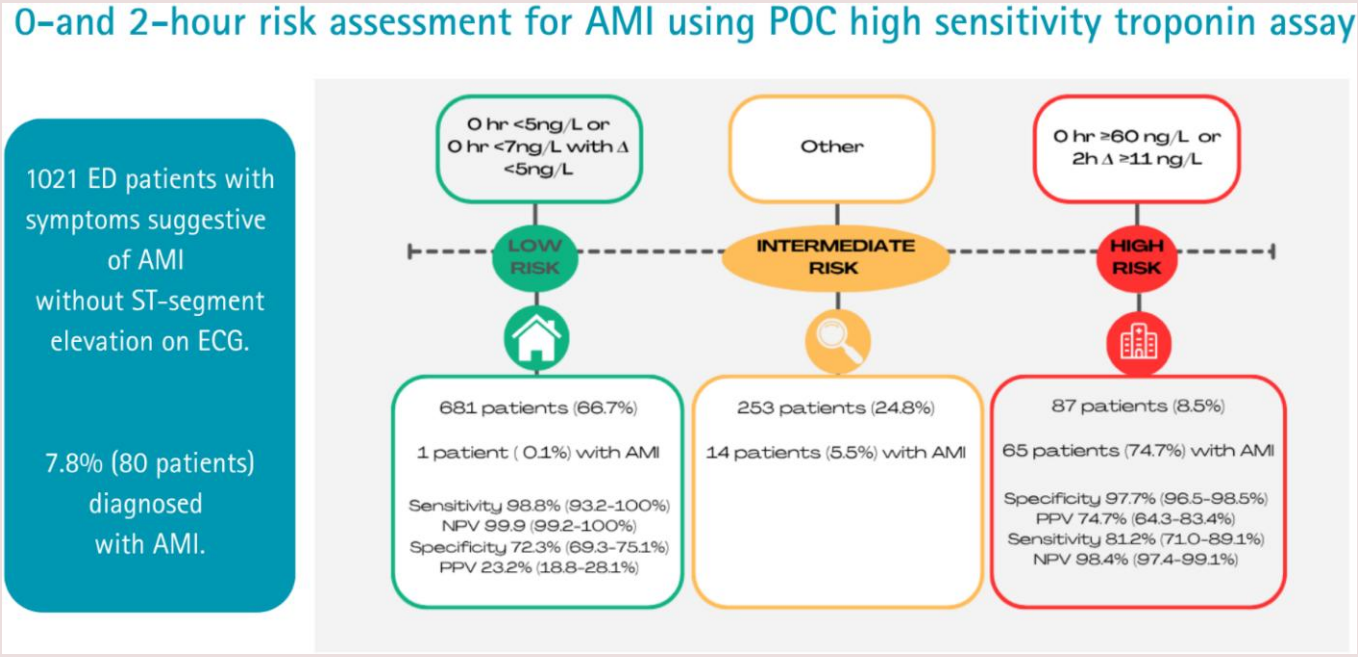
Graphical Abstract



AMI, acute myocardial infarction; POC, point of care; ED, emergency department; ECG, electrocardiogram; NPV, negative predictive value; PPV, positive predictive value

Keywords Troponin • Point-of-care • Acute myocardial infarction • Emergency department

Introduction

Cardiac troponin (cTn) is critical for the evaluation of the large numbers of patients who present to emergency departments (EDs) with suspected acute coronary syndrome (ACS), which represents between 5 and 10% of all adult presentations.[1] The finding of elevated cTn values, supportive of myocardial injury, along with ECG and other clinical information suggesting ischaemia, are used to diagnose acute myocardial infarction (AMI).[2] High sensitivity cTn (hs-cTn) assays have improved the safety and management of emergency patients with suspected ACS.[3,4] In practice, hs-cTn assays with improved analytical sensitivity have been shown to reduce overall healthcare costs and patient mortality.[5]

Until recently, hs-cTn assays were only available on laboratory-based platforms, with ideal turnaround time (TAT) for results of <60 min, but typically longer.[6] Although accurate, the timeliness of laboratory-based assay may result in delays in decision-making, patient care, and disposition. New point-of-care (POC) hs-cTn platforms allow near-patient testing and significantly reduced time to results to <20 min.[7] Understanding the performance of new assays with well-validated central-laboratory assays is valuable. Optimum metrics for clinical use of each new troponin assay are required to be defined, which supports the safe risk stratification of emergency patients with suspected ACS.[7] This study aimed to derive a 0 and 2 h strategy for the risk assessment of patients with suspected AMI to the ED using the Abbott i-STAT® hs-TnI assay, a new POC assay, and to compare results with those from two laboratory-based assay.

Methods

Study design and setting

This prospective observational trial incorporates data from the Suspected Acute Myocardial Infarction in Emergency (SAMIE) trial (ACTRN12621000053820). This study recruited patients from five Australian hospitals between 17 November 2020 and 9 September 2021. HREC approval was obtained (LNR/2020/QRBW/65773) and written informed consent obtained from all patients.

Eligible patients were those aged ≥18 years of age and the treating physician intended to investigate for AMI. Exclusion criteria included ST-segment elevation myocardial infarction (STEMI) on ECG interpretation, transfer from another hospital, previous enrolment in the trial within the past 30 days, or inability or unwillingness to provide informed consent. Eligible patients were recruited between 7:30 a.m. and 4:00 p.m., Monday to Friday.

All patients were managed in accordance with the local standard care, including single ECG and troponin testing for a subset of patients with a troponin concentration below the limit of detection (LoD) at presentation and serial testing at 2–3 h for all other patients. Additional testing for ischaemic heart disease, such as CT coronary angiography, was at the discretion of the treating clinician.

Data collection and processing

Standardized case report forms were used. Baseline characteristics, risk factors, and past medical history were gathered from the patient record by dedicated research nurses.[8,9]

Blood samples were collected at presentation (0 h) and 2 h later for both POC hs-cTnI testing and central laboratory reference testing. The troponin assay used in clinical care at all sites was the Beckman Coulter Access 2 [limit

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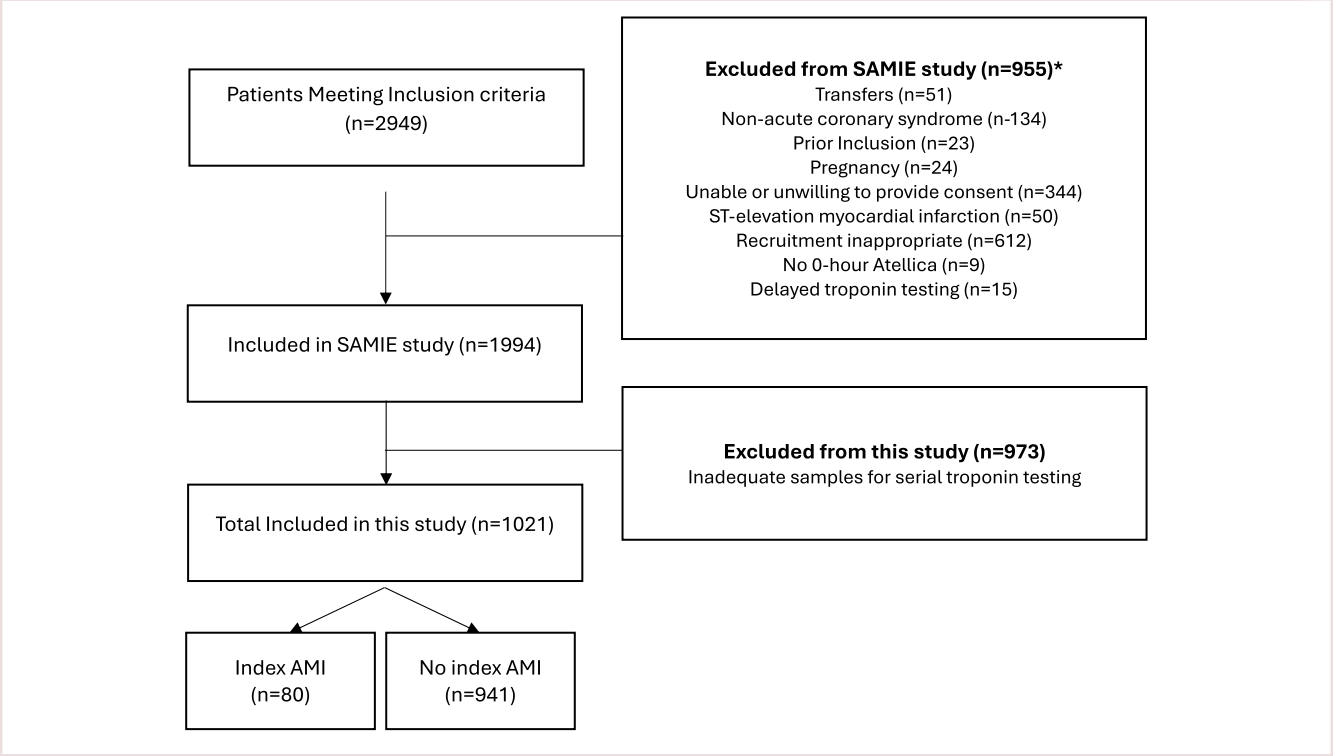

Figure 1 Patient flow diagram. * indicates that patients could have multiple exclusion criteria. SAMIE, suspected acute myocardial infarction in the emergency department; AMI, acute myocardial infarction.

of detection (LoD) 2.3 ng/L; 99th percentile upper reference limits (URLs) of 11.6 ng/L and 19.8 ng/L for females and males, respectively].[10] Clinical cut points used for females and males were 10 and 20 ng/L, respectively, and LoD of 2 ng/L.

An additional 10 mL of blood was collected from patients during the routine 0 and 2 h blood draws. These lithium-heparin plasma samples were sent to the local central laboratory. Some of each sample was used to test the Siemens Atellica laboratory assay (LoD 1.6, 38.6, and 53.5 ng/L female and male URLs, respectively[10]) and the rest was centrifuged for 10 min at 3000 *g*, aliquoted, and frozen to −80°C. Samples were later thawed and tested on the Abbott i-STAT® hs-TnI assay at a local private pathology provider by trained laboratory staff. This testing occurred between December 2024 and April 2025 (4 years after collection). In the event of a processing error, the same sample underwent a repeat test. The investigational POC assay was the Abbott i-STAT® hs-TnI assay, with an overall 99th percentile in whole blood of 21 ng/L and sex-specific 99th percentiles of 13 ng/L for females 28 ng/L for males. The assay is a high-sensitivity troponin assay with the LoD being 1.6ng/L for whole blood and 1.05nl/L for plasma and the 10% CV at 6.89 ng/L for whole blood and 3.7 ng/L for plasma.[10,11]

Thirty-day follow-up was conducted by research nurses, which included direct telephone contact to determine if further cardiac testing or hospital admission had occurred. Statewide databases were interrogated to identify if further cTn testing or admission to a public hospital had occurred. If relevant, additional clinical information was obtained, including revascularization details.

Endpoint adjudication

Senior clinicians reviewed all cases where at least one cTn concentration was above the sex-specific 99th percentile based on the assay in clinical use (Beckman Coulter hs-cTnI assay), blinded to the POC assay result. Adjudication was based on all available information up to at least 30 days after the initial presentation, including 12-lead ECGs, cardiac investigations and procedures, Beckman Coulter hs-hs-cTnI concentrations, and clinical records. All patients in the current study had serial troponin results. Second review occurred at clinician request in complex cases.

The diagnosis of MI was defined according to the Fourth Universal Definition of MI (UDMI) and required a rise and/or fall in cTn with at least one concentration >99th percentile in association with acute ischaemic symptoms, development of pathological Q waves in the 12-lead ECG, ECG changes indicative of new ischaemia, imaging evidence of new loss of viable myocardium or new regional wall motion abnormality, or identification of an intracoronary thrombus by angiography or autopsy.[2]

Patients with an adjudicated diagnosis were further classified as type 1 or type 2 MI (T1MI; T2MI) in accordance with the UDMI.[2] T1MI required evidence of atherosclerotic plaque disruption. T2MI required imbalance between myocardial oxygen supply and demand not due to atherothrombosis. Patients not meeting these criteria with elevated troponin concentrations were defined as having myocardial injury.

Study outcomes

The primary composite endpoint was AMI (Index AMI) or cardiac death during the index presentation, including T1MI (NonSTEMI; NSTEMI), STEMI occurring after ED presentation, T2MI, or cardiac death (due to MI, unstable angina, coronary artery disease, congestive cardiac failure, and cardiac arrhythmia). The secondary endpoints were MI (T1 and T2; STEMI, and NSTEMI) and MACE within 30 days. MACE included urgent revascularization and cardiac death (including index admission).

Data analysis

Continuous baseline variables are presented as mean (standard deviation) or median (interquartile range), while categorical variables are presented as number (%). These data are reported by study cohort and AMI status. Analyses were performed using Stata version 17. Coding is available on request from the authors.

The area under the receiver operating characteristic (AUROC) curve was calculated for the Abbott POC assay, the Siemens Atellica laboratory assay, and the Beckman Access assay, which was considered the gold standard as it was used as part of standard patient management.

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The derived algorithm included a single 'very low-risk' cTnI value, chosen based on previous literature,[11,12] and a 'low-risk' cTnI value with a 2 h delta. The selection of these parameters for the low-risk cohort was a combination of values that achieved a negative predictive value (NPV) ≥99.5% and sensitivity of ≥99% for the primary endpoint. The algorithm also included a 'high-risk' cTnI value or delta. We aimed to identify values that would achieve a PPV >70% while maximizing the proportion of patients that would be ruled in. Multiple cutoffs achieved equivalent diagnostic accuracy. Therefore, the final cutoff was selected through bootstrapped sampling. Diagnostic accuracy statistics were calculated in 100 bootstrapped samples for each candidate cutoff. Values that achieved PPV >70% in >80% of samples were considered, and the cutoff that maximized the proportion of patients ruled in across all samples was chosen.

Diagnostic accuracy statistics were reported for the primary (index AMI and cardiac death) and secondary endpoints (index T1MI and 30 day MACE). Sensitivity analyses also were conducted where diagnostic accuracy statistics were reported by subgroups. These included patients presenting <2 h or ≥2 h (early vs. late presenters) from symptom onset and by sex. The performance of the algorithm was validated using a bootstrap procedure. Specifically, diagnostic accuracy statistics were computed in 100 bootstrap samples, and descriptive statistics on these samples were reported.

Results

Overall, 1021 participants were included. As show in in *Figure 1*, patients were excluded from the current analysis if they had with inadequate samples for serial POC testing ($n = 973$). All participants included in the study had serial hs-cTnI testing. Eighty participants (7.8%) had an adjudicated diagnosis of index AMI, including 6 (0.6%) with STEMI after presentation, 55 (5.4%) with T1 NSTEMI, and 19 (1.9%) with T2 AMI. In subjects who met the diagnosis of myocardial infarction a type 1 MI (STEMI and NSTEMI) was diagnosed when confirmed by angiography during the index admission in 44 of 61 cases. Where angiography was not performed, type 1 MI was diagnosed where AMI occurred spontaneously and without any obvious precipitant for a type 2 AMI. There were no cardiac deaths during index admission.

The mean age of the total population was 58.5 years (SD 14.4 years), with 56.5% male participants. A total of 749 (73.5%) of patients presented >2 h after symptom onset. Baseline characteristics of patients by AMI status are shown in *Table 1*.

The area under the receiver operating characteristic curve (AUROC) for the primary outcome was calculated based on the highest troponin value within 2–3 h of presentation. This was 0.98 (95% CI: 0.96–0.99) for the Abbott i-STAT® hs-TnI POC assay, 0.97 (95% CI: 0.95–0.99) for the Siemens Atellica assay, and 0.98 (95% CI: 0.98–0.99) for the Beckman Coulter assay (*Figure 2*). Correlations were high between the Abbott i-STAT® hs-TnI POC assay and the Beckman Coulter assay at 0 h (Spearman's rho = 0.88, $P < 0.01$) and 2 h (Spearman's rho = 0.89, $P < 0.01$. Correlations were also high between the Abbott i-STAT® hs-TnI POC assay and the Atellica assay at 0 h (Spearman's rho = 0.91, $P < 0.01$) and 2 h (Spearman's rho = 0.92, $P < 0.01$).

Risk stratification performance: derivation

Diagnostic accuracy for the derived strategy is shown in *Figure 3*. The initial value of either <5 or <7 ng/L with delta <5 ng/L over 2 h identified 681 (66.7%) patients as low risk for AMI and achieved an NPV of 99.9% (95% CI: 99.2–100%) with sensitivity of 98.8% (95% CI: 93.2–100%). Of these, there were 547 (53.6%) patients deemed low risk with a 0 h value of <5 ng/L and 134 (13.1%) deemed low risk with an initial value <7 ng/L and delta <5 ng/L over 2 h.

High-risk criteria were met in 87 patients (8.5%), with 65 true positives [74.7% positive predictive value (PPV) in the high-risk group]. Of these, 58 (5.7%) were deemed high risk based on an initial value of ≥60 ng/L and 29 (2.8%) based on a delta ≥11 ng/L. The remaining 253 patients (24.8%) fell into an intermediate zone requiring additional evaluation.

Table 1 Patient characteristics

Characteristics	No index AMI ($n = 941$)	Index AMI ($n = 80$)	*P*
Mean age, years (SD)	57.9 (14.4)	66.3 (12.1)	<0.01
Male sex, *n* (%)	519 (55.2%)	58 (72.5%)	0.01
Aboriginal and/or Torres Strait Islander, *n* (%)	45 (4.8%)	1 (1.2%)	0.14
Chest pain, *n* (%)	889 (94.5%)	77 (96.2%)	0.78
Present >2 h after symptom onset, *n* (%)	688 (73.3%)	61 (76.2%)	0.56
Medical history, *n* (%)			
AMI	185 (19.7%)	32 (40.0%)	<0.01
CABG	64 (6.8%)	11 (13.8%)	0.07
Angioplasty	166 (17.6%)	24 (30.0%)	0.01
CAD	239 (25.4%)	34 (42.5%)	<0.01
Risk factors, *n* (%)			
Hypertension	479 (50.9%)	51 (63.7%)	0.03
Diabetes	178 (18.9%)	24 (30.0%)	0.02
Dyslipidaemia	456 (48.5%)	54 (67.5%)	<0.01
Family history of CAD	409 (43.5%)	30 (37.5%)	0.30
Smoking	213 (22.6%)	19 (23.8%)	0.82
Median BMI (IQR)	30.8 (7.6)	29.2 (6.4)	0.15
Median time to first troponin (IQR)	0.7 (0.4–1.0)	0.6 (0.4–0.9)	0.11
Median ED LOS, hours (IQR)	6.1 (5.2–7.4)	6.1 (4.0–8.1)	0.21
Mean hospital LOS, hours (IQR)	6.7 (5.6–19.6)	74.1 (40.6–134.8)	<0.01

AMI, acute myocardial infarction; SD, standard deviation; CABG, coronary artery bypass graft; CAD, coronary artery disease; BMI, body mass index; IQR, interquartile range.
There were missing data for time of presentation ($n = 2$) and BMI ($n = 396$).

Risk stratification performance: internal validation

The diagnostic accuracy of the algorithm was tested on 100 bootstrapped samples. Sensitivity for low-risk cutoffs ranged from 95.2 to 100% (mean 98.7%). Ninety-five per cent of samples had a sensitivity ≥96.4%. NPV ranged from 99.4% to 100% (mean 99.8%). Ninety-five per cent of samples had an NPV ≥99.6%.

The high-risk criteria yielded specificity from 96.6 to 98.7% (mean 97.7%). Ninety-five per cent of samples had a specificity ≥96.9%. PPV ranged from 63.4 to 85.7% (mean 75.7%). Ninety-five per cent of samples had an NPV ≥67.8%.

Safety outcomes

One patient in the low-risk group experienced a T1 NSTEMI during the index presentation. This was a 46-year-old male with Abbot POC troponins of 3 and 4 ng/L at 0 and 2–3 h. This patient had hypertension, diabetes, dyslipidaemia, and a family history of CAD. His

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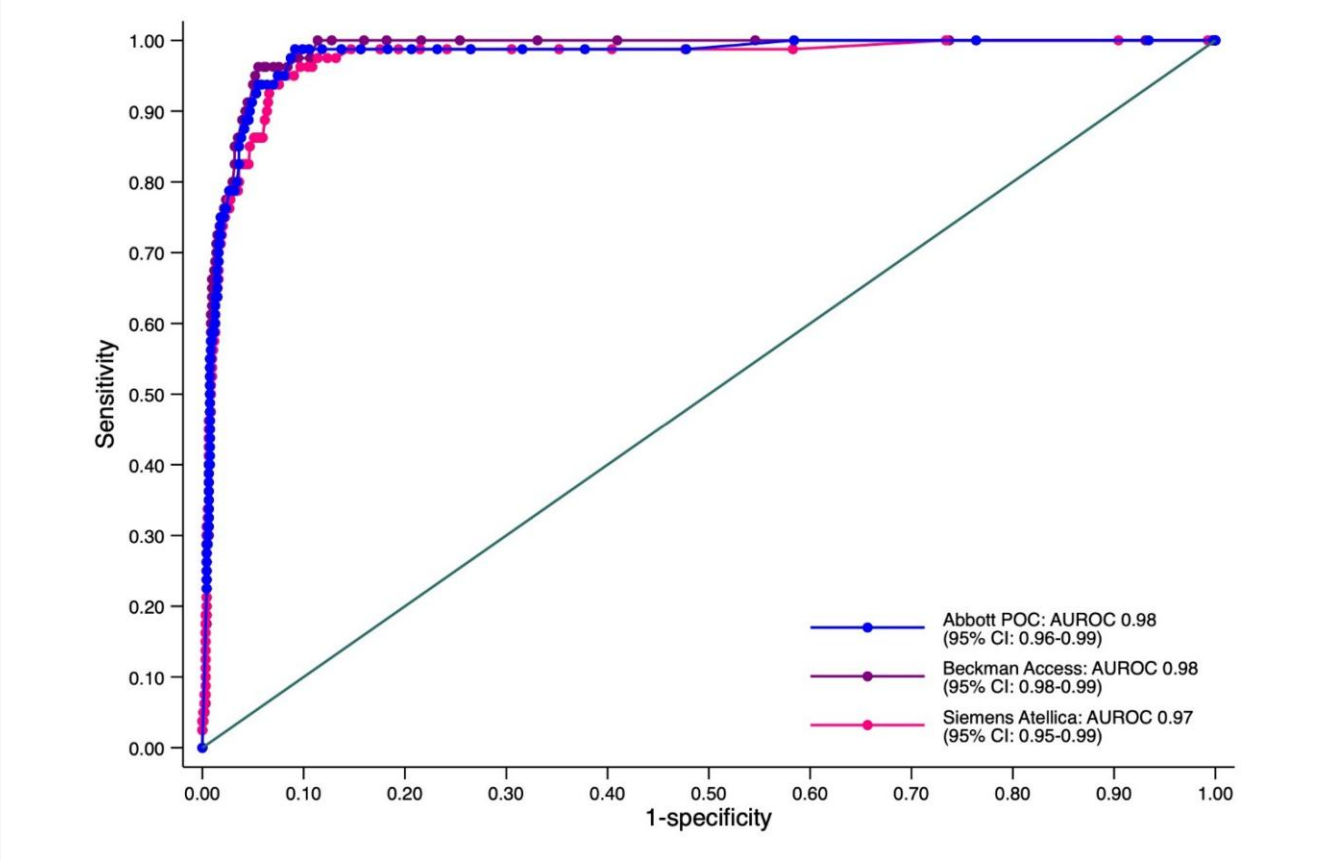

Figure 2 Receiver-operating characteristic curves for the primary outcome describing the maximum troponin concentration over 2–3 h using three high sensitivity cardiac troponin assays. Abbott i-STAT® hs-TnI POC, Siemens Atellica and Beckman Coulter hs-cTnI assays. POC, point of care; AUROC, area under the receiver operator curve; CI, confidence intervals.

highest cTn concentration reported by the Beckman Coulter hs-cTnI assay was 28 ng/L and he was found to have triple vessel disease on CTCA.

Secondary/sensitivity analyses

There were 61 patients with index T1MI and 86 (8.4%) with 30 day MACE (*Table 2*). An additional five patients in the low-risk group were classified as having a MACE by 30 days (see Supplementary material online, *Table S1*). Four of these presented with unstable angina and underwent revascularization within 30 days. The fifth patient presented with unstable angina and was discharged after cardiology review to undergo outpatient CTCA. This patient re-presented one day later with a troponin of 29 ng/L (Beckman Coulter) and underwent PCI for T1 NSTEMI.

There were 270 (26.4%) patients presenting within 2 h of symptom onset, 749 (73.4.9%) presenting after 2 h, and 2 (0.2%) with an unknown symptom onset time. Diagnostic accuracy statistics for early and late presenters are provided in Supplementary material online, *Table S2*. For patients presenting <2 h after symptom onset, no patients had the primary endpoint (sensitivity 100%, 95% CI: 82.4–100% and NPV 100%, 95% CI: 98.0–100%). For patients presenting ≥2 h after symptom onset, there was one missed NSTEMI (sensitivity 98.4%, 95% CI: 91.2–100 and NPV 99.8%, 95% CI: 98.9–100%). Index AMIs occurred for 4.6% of females and 10.1% of males. Sensitivity for AMI was 100% (95% CI: 85.6–100%) for females and 98.3% (95% CI: 90.8–100%) for males.

Discussion

This large study derived a 0 and 2 h POC strategy using the Abbott i-STAT® hs-TnI POC assay to risk stratify emergency patients with suspected AMI. Although the data on the 0 h metrics have been reported, the combination of 0 and 2 h serial testing for risk assessment has not.[11] Combining the analytical sensitivity of the hs-cTn assay with the rapid turnaround time of the POC platform (15 min) has the potential to accelerate decision-making and may ease overcrowding in the ED setting with disposition planning possible in over two-thirds of patients (66.7%) with 2 h results.

The study provides a safe and accurate protocol for the identification of patients at low risk for AMI and MACE, supporting rapid, safe discharge from the ED of 66.7% of patients within 2 h of testing. Using a cutoff of <5 ng/L would enable 53.6% to be identified as low risk with the 0 h results alone. With serial measurements, including those with values initially of <7 ng/L with a change over 2 h of <5 ng/L, a further 13.1% would be identified with an NPV of 99.8% (95%CI 99.2–100%). Our low-risk criteria are in keeping with international and national guidelines recommending that patients deemed low-risk (<1%) may be discharged from the ED.[13,14]

The proportion of patients identified as low risk is in keeping with other guideline-supported strategies that endorse prompt discharge of low-risk patients.[14,15] For example, using other hs-cTn POC assays, the Triage True assay identifies 45%, PATHFAST up to 38% and Atellica VTLi POC assay up to 42% patients to be deemed safe for discharge with 0 h results.[9,16,17] This study builds upon the knowledge of the metric using

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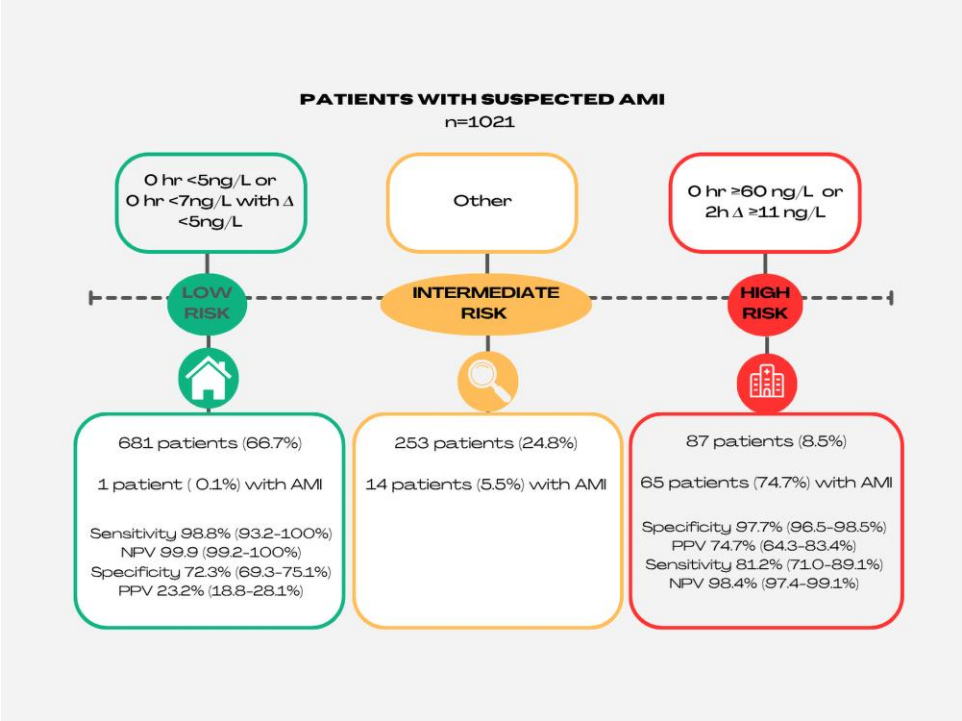

Figure 3 Diagnostic accuracy for the derived 0/2 hour strategy. AMI, acute myocardial infarction; NPV, negative predictive value; PPV, positive predictive value.

Table 2 Secondary endpoints

	LOW risk 681 patients (66.7%)	INTERMEDIATE risk 253 patients (24.7%)	HIGH risk 87 patients (8.5%)
T1MI	1 (0.1%) with T1MI Sensitivity: 98.4% (95% CI: 91.2–100%) Specificity 70.8% (95% CI: 67.8–73.7%) NPV: 99.9% (95% CI: 99.2–100%) PPV: 17.6% (95% CI: 13.7 -22.1%)	9 (3.6%) with T1MI (3.6%, 95% CI: 1.6–6.6%)	51 (58.6%) with T1MI Sensitivity: 83.6% (95% CI: 71.9–91.8%) Specificity 96.2% (95% CI: 94.8–97.4%) NPV: 98.9% (95% CI: 98.0–99.5%) PPV: 58.6% (95% CI: 47.6–69.1%)
30 day-MACE	6 with 30 day MACE Sensitivity 93.0% (95% CI: 84.5–97.4%) Specificity: 72.2% (95% CI: 69.2–75.0%) NPV: 99.1% (95% CI: 98.1–99.7%) PPV: 23.5 (95% CI: 19.1–28.4%)	15 with 30 day MACE (5.9%, 95% CI: 3.4–9.6%)	65 with 30 day MACE Sensitivity 75.6% (95% CI: 65.1–84.2%) Specificity 97.6% (95% CI: 96.5–98.5%) NPV: 97.8% (95% CI: 96.6–98.6%) PPV 74.7% (95% CI: 64.3–83.4%)

MACE, major adverse cardiac events; T1MI, Type 1 myocardial infarction; NPV, negative predictive value; PPV, positive predictive value.

the Abbott i-STAT® hs-TnI POC assay on presentation to define low risk patients (5 ng/L), improving the confidence in accuracy of this value due to the larger cohort investigated.[11] Additionally, early identification of patients at high risk for MI enables rapid referral for admission. Similar risk stratification processes have already been shown to improve the efficiency of ED assessment; however, most reported laboratory-based assay results.[3,18,19] Incorporation of POC results may facilitate even earlier (or timelier) decision-making, adding to these benefits.

The maximum Abbott POC i-STAT hs-TnI concentrations showed high correlation with two laboratory-based hs-cTnI assay (Abbott and Siemens) results over 2–3 h post ED presentation. Other POC assays have been evaluated and compared with laboratory-based assays, including the POC Triage system (QuidelOrtho; San Diego, USA), PATHFAST cTnI-II assay (LSI Medience Corporation; Mitsubishi Chemical Europe, Dusseldorf, Germany), and Atellica VTLi (Siemens Healthineers; Forchheim, Germany).[9,16,17] Our study adds to the growing knowledge about the accuracy and use of such near-patient diagnostics in comparison to laboratory-based assays. Accuracy is essential to support use of POC devices and other near-patient diagnostics with results rapidly available (<20 min) to allow early decision-making by clinicians and thereby improving ED efficiency and reducing ED overcrowding.

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The portability and ease of use of POC devices may also support assessment of patients with suspected ACS in non-hospital environments such as general practice, rural and regional health centres, and ambulance settings where access to laboratory-based pathology services may be limited. Further research is needed to ensure that adoption of these strategies is safe in these environments, given the likely earlier presentation and therefore sampling in some situations (e.g. ambulance).[7]

Limitations

Several limitations should be acknowledged. The AMI prevalence of 7.8% may not reflect all ED settings and the performance of the protocol in higher risk (e.g. cardiology wards) or lower risk (e.g. general practice settings) cannot be assumed to be equivalent. This is a derivation study only, and an external validation is required. The results were obtained from a prospective study using stored plasma samples. Plasma and whole blood comparison studies show results from these sample types are equivalent but not identical. The use of whole blood may report slightly higher values than plasma, which would lower estimates of sensitivity and increase estimates of specificity when using plasma. Validation of this strategy using whole blood is required to validate the cutoffs. Analysis of blood was performed in a laboratory setting rather than by clinical staff in a clinical setting. Further research in a clinical setting is required. The coefficients of variation obtained in the testing laboratory within the range of the decision limits are not known. Values reported are specific to this assay and should not be used with other hs-cTn assays. As the diagnosis of AMI required the finding of an elevated troponin value, the outcomes reported were influenced by the clinical assay in use. The investigational assay therefore can never be better than this assay. The fourth UDMI was used by trained cardiologists to adjudicate outcomes. In most instances only one cardiologist was involved. Not all subjects underwent diagnostic angiography during the index admission; therefore, it is possible that some cases of type 2 MI (e.g. due to non-atherosclerotic coronary dissection or coronary embolus) could have been misclassified as type 1 MI. Misclassification cannot be excluded. Patient recruitment only occurred between 7:30 a.m. and 4:00 p.m., and the extent of selection bias cannot be determined. We have previously reported no differences in demographics or outcomes of patients presenting within or out of these hours.[20] This trial was an observational derivation study only. A prospective validation and implementation trial would be needed to determine the impact on patient care and health services.

The Abbott POC i-STAT hs-TnI assay provides safe risk stratification of emergency patients with suspected AMI. Its clinical performance is comparable to two laboratory-based assays. The reported protocol with results within 15 min may facilitate early decision-making supporting patient care and health service efficiency and should be evaluated in a clinical validation study.

Supplementary material

Supplementary material is available at *European Heart Journal: Acute Cardiovascular Care* online.

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Data availability

Access to de-identified data may be granted by the corresponding author upon reasonable request and with approval from the relevant Human Research Ethics Committee(s).

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